

$$\|u+v\|^2 = \langle u+v, u+v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle$$

$$\|u-v\|^2 = \langle u-v, u-v \rangle = \langle u, u \rangle - \langle u, v \rangle - \langle v, u \rangle + \langle v, v \rangle$$

$$\|u+v\|^2 - \|u-v\|^2 = 2\langle u, v \rangle + 2\langle v, u \rangle = 2\langle u, v \rangle + 2\overline{\langle u, v \rangle} = 4 \operatorname{Re} \langle u, v \rangle$$

$$\operatorname{Re} \langle u, v \rangle = \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2)$$

$$\|u-iv\|^2 = \langle u-iv, u-iv \rangle = \langle u, u-iv \rangle - i\langle v, u-iv \rangle =$$

$$= \langle u, u \rangle + i\langle u, v \rangle - i\langle v, u \rangle + \langle v, v \rangle$$

$$\|u+iv\|^2 = \langle u+iv, u+iv \rangle = \langle u, u \rangle - i\langle u, v \rangle + i\langle v, u \rangle + \langle v, v \rangle$$

$$\|u-iv\|^2 - \|u+iv\|^2 = 2i\langle u, v \rangle - 2i\langle v, u \rangle = 2i\langle u, v \rangle - 2i\overline{\langle u, v \rangle}$$

$$2i(u-v) = 2i \cdot 2i \operatorname{Im}(u) =$$

$$= -4 \operatorname{Im} \langle u, v \rangle$$

$$\operatorname{Im} \langle u, v \rangle = \frac{1}{4} (\|u+iv\|^2 - \|u-iv\|^2)$$